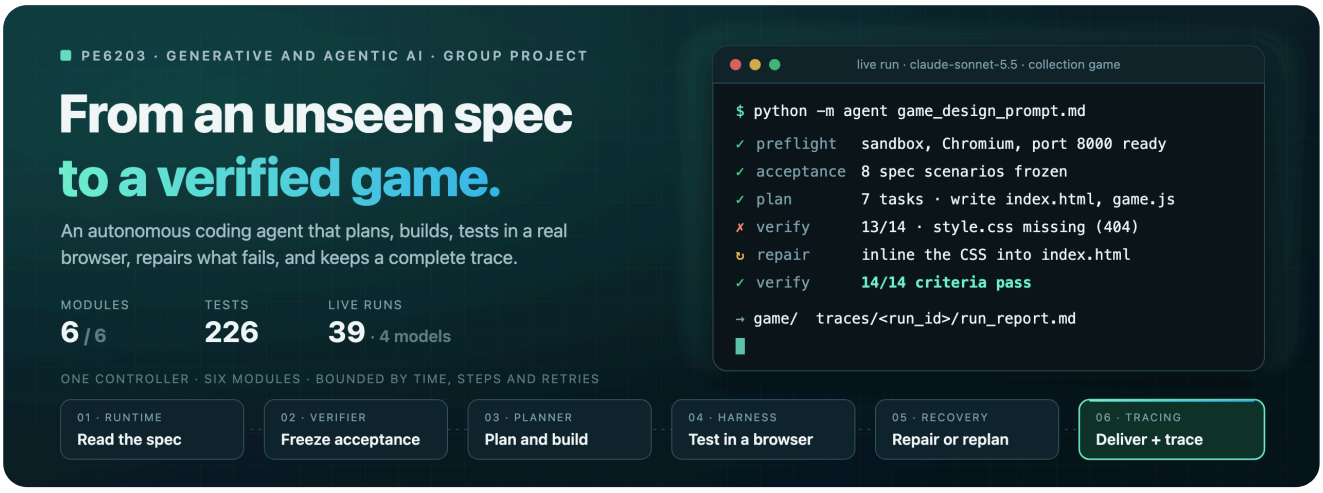
PE6203 · GENERATIVE AND AGENTIC AI · GROUP PROJECT 1

A Coding Agent That Builds a Game

Project report: architecture, planning and verification, failure recovery, evaluation, limitations and next steps



Member	Name	Module owned
1	ZHOU YU	Controller, CLI, configuration, team contract, integration
2	JIN ZIYI	Planner: model client, prompts, plan and decision parsing
3	DAI MINFEI	Tool runtime: files, commands, sandbox, processes, dependencies
4	ZHOU SIHAN	Verifier: acceptance criteria and the Playwright game harness
5	LIN SIYUAN	Failure recovery policy and offline evaluation
6	LI JIAKUN	Tracing, run report and delivery audit

Repository: github.com/shirley830/6203-game-coding-agent · Entry point: `python -m agent game_design_prompt.md`

Instructor: Prof. Bo An · Teaching assistant: Feng Lang · October 2026

1 Summary

We built an autonomous coding agent that starts from a single, previously unseen `game_design_prompt.md`, writes a complete game into `game/`, and leaves a full execution record in `traces/`. The agent is started with one documented command and needs no human help afterwards. Its distinguishing design choice is that **success is never declared by the language model**: a deterministic controller grants success only when every mandatory acceptance criterion passes on the current code, with evidence, and those criteria are derived from the specification *before* any code exists and checked by actually playing the game in a headless browser.

6 / 7	39	226	6
specifications passed in the final GLM-5.3 rehearsal	autonomous runs with four real models	automated tests, all passing	modules behind one controller and one contract

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2 System architecture and rationale

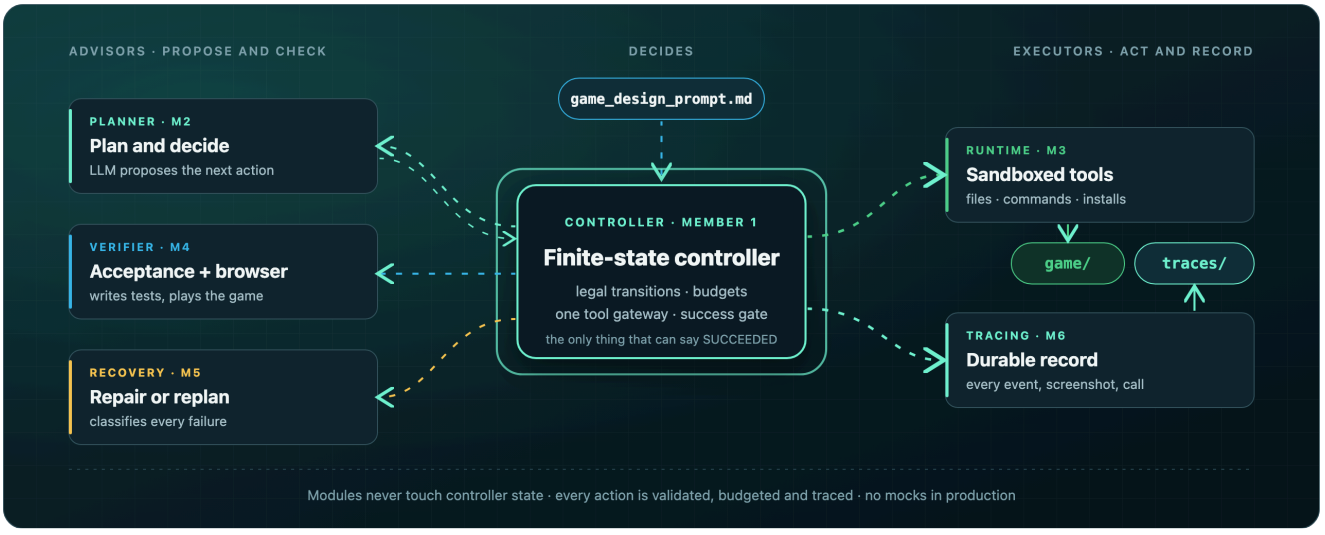


Figure 1. One finite-state controller decides. The planner, verifier and recovery modules propose and check; the sandboxed runtime and tracing act and record.

The agent is a **single sequential controller** (Member 1) with five injected modules. Modules receive an immutable view of the run and return a typed proposal or observation; they cannot change controller state, call each other, or run tools themselves. Every file write, command and dependency installation

passes through one tool gateway in the controller, which validates its scope, charges it to the budgets, records it in the trace and only then hands it to the runtime. All modules are wired by one production factory (`agent/integrations.py`); test doubles exist only under `tests/`.

Module	Contract	Responsibility
Controller (M1)	state machine, budgets	Phases, legal transitions, tool gateway, stopping conditions, the success gate
Planner (M2)	`create_plan`, `next_decision`	Plan from frozen criteria; propose exactly one next action per call
Runtime (M3)	`execute`	Files, commands, installs inside an OS sandbox; process-group cleanup; workspace revisions
Verifier (M4)	`prepare_acceptance`, `next_check`	Spec-derived acceptance; read-only checks; browser play-testing; evidence
Recovery (M5)	`recommend_recovery`	Classify each failure; recommend RETRY, REPAIR, REPLAN or ABORT
Tracing (M6)	`open_run`, `append_event`, `finalize`	Durable events, checkpoints, evidence, run report, delivery check

Design rationale and trade-offs

- **One controller instead of a multi-agent system.** A single owner of state makes every decision traceable and every budget enforceable, and it removes coordination failures between agents. The cost is less parallelism; for one game in a bounded time slot, predictability mattered more than throughput.
- **The model proposes, the controller decides.** Language models are good at generating code and poor at judging their own completion. Success therefore requires a verification report that matches the current specification digest, workspace revision and acceptance-suite version, with evidence for every mandatory criterion.
- **Typed contracts first.** The team agreed a written interface contract (`docs/INTERFACES.md`) before implementation, so six members could build modules in parallel and integrate them without changing each other's code.
- **Browser games by default.** A self-contained HTML5 canvas game needs no build step or installation and can be verified by a real browser. The agent builds Python terminal or pygame games only when the specification asks for them, as the course allows both.
- **Provider independence.** The model is reached through LiteLLM, so changing to the course model required only configuration. Reasoning models needed one provider-specific setting (Section 7.3).

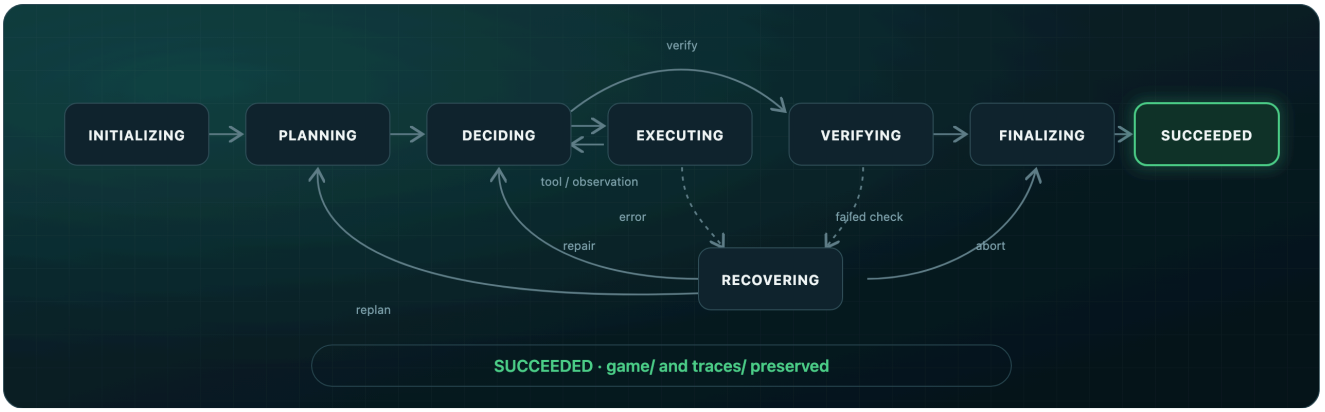


Figure 2. The run state machine. Errors, failed checks and uncertain outcomes go to RECOVERING, which repairs (back to DECIDING), replans (back to PLANNING) or aborts. Budget exhaustion or cancellation finalizes from any phase; finalization always preserves `game/` and `traces/`.

3 How the agent plans

Plan. After the acceptance suite is frozen (Section 4), the planner asks the model for 3-8 concrete tasks. Each task must reference acceptance criteria; the plan must cover every mandatory criterion and have an acyclic dependency graph, otherwise it is rejected and regenerated. The plan prompt treats the specification as data that cannot override system policy.

Decide, one action at a time. In the build loop the planner returns exactly one decision per call: a tool action (write, patch, read, inspect, run, install), a verification request with a launch command, or a plan revision. Because the controller exposes only the latest result, the planner keeps its own **working memory**: the files in `game/`, the last 25 actions and failures, guard notes, and facts from the start-up self-check (for example "node is not installed"). This keeps prompts small while preserving context across steps.

Guards. Some rules are enforced in code rather than trusted to the model:

- A verification request is refused if the launch command refers to files that do not exist; `game/README.md` must contain the exact launch command and is written deterministically if missing.
- After eight consecutive edits without verification, the planner requests verification itself. Live runs showed a model rewriting one file 35-45 times without fresh feedback until the time budget ran out.
- The planner owns action IDs (models reuse example IDs) and repairs unambiguous format slips, such as `{"decision": "read_file"}` instead of a tool action of kind `read_file`. Calls use JSON mode.
- **Specification fidelity.** The prompt forbids adding controls, shortcuts or behaviour outside the specification to make a check pass, and requires fixing the real defect instead (Section 4.4).

4 How the agent verifies its work

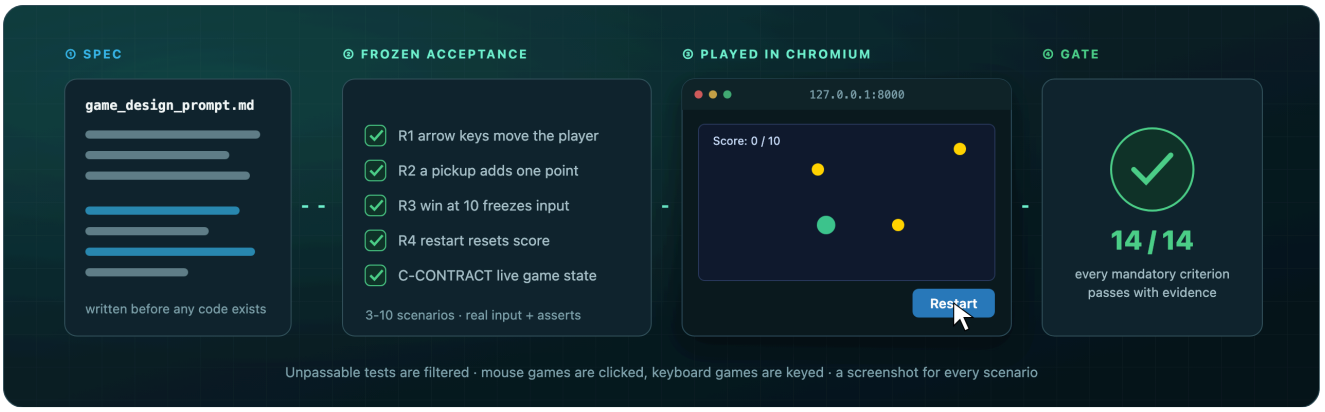


Figure 3. Verification: the specification becomes frozen executable scenarios before any code exists; the harness plays the game in headless Chromium; failures return to the planner with the failing step and the game state.

4.1 Acceptance is frozen before planning

Immediately after reading the specification, the verifier asks the model, in the role of an independent tester, for a **state contract** and **3-10 executable scenarios**. The contract requires the game to expose its real live state as `window.__gameState` and a few setup hooks as `window.__gameTest` (for example placing a collectible next to the player). A scenario is a list of real inputs (key presses, held keys, clicks on DOM elements or grid cells) and JavaScript assertions over the state. These criteria are frozen with the generic launch, functional, interaction and technical criteria; neither planning nor repair can change them.

Model-written tests can be wrong, so each requirement is validated on its own: invalid steps, undeclared hooks, or canvas clicks in a specification without mouse input drop that requirement only. If fewer than three survive, or the call fails, the verifier falls back to the generic criteria and records why.

4.2 The game is actually played

The harness starts the documented launch command, loads the page in headless Chromium and records JavaScript errors. It checks the contract on the live object, then runs every scenario on a freshly loaded page, saving a screenshot per scenario. A failing scenario reports the step that failed and the state after it, for example `step 5 {"expect": "s.status === 'lost'"} … state after: {...}`, which is exactly what the planner needs to repair the game. A generic interaction probe (keys, canvas points, board cells, buttons) and checks for a play surface complement the scenarios. Terminal and pygame games are verified in a process mode: start without a traceback, keep running, react to typed moves (pygame input can only be checked statically and is labelled as such).

4.3 The success gate

The controller computes the outcome; a verifier cannot simply claim success. A report passes only if its specification digest, workspace revision and acceptance-suite version are current, every criterion has a result, every mandatory result is PASS with non-empty evidence, at least one verification tool operation succeeded in this cycle, a launch command exists, and the trace is durably preserved.

4.4 Why this matters: evidence from live runs

Observation	Consequence
With generic checks only, all six first-round runs "succeeded", but manual review found specification violations in three of them (for example, walking into the enemy did not lose).	Spec-derived scenarios now catch these: the same "enemy contact loses" rule failed in every cycle on the faulty game, and GLM-5.3 fixed it after one failed cycle.
When the harness could only press keys, the agent added keyboard controls and click-anywhere behaviour to a mouse-only memory game to make the probe pass.	Mouse-aware scenarios and the fidelity rule removed this: later runs passed by real clicks with no extra controls.

5 How the agent recovers from failure

Every failure is recorded with its phase and category and goes to the recovery module, which classifies it and recommends one of four decisions. The controller enforces the decision and all limits; modules cannot reset counters.

Failure class	Decision	Reasoning
Transient model error (timeout, connection, rate limit, 5xx)	RETRY	Proposal-only calls are safe to repeat; waits 5 s, 15 s, 45 s, 60 s before each retry, within the deadline
Malformed model output	REPAIR / REPLAN	Ask again with the parse error visible; an invalid plan is regenerated
Command failed, file missing	REPAIR	Inspect output and fix; never replay the unchanged command
Timeout or unknown side effects	REPAIR	**Never retried blindly**; earlier verification is invalidated and the workspace re-inspected
Failed verification	REPAIR, periodically REPLAN	The failing steps go to the planner; repeated failure rebuilds the plan
Permission, sandbox or credential error; process cleanup failure	ABORT	Retrying cannot help; preserve artifacts and finalize honestly

Stopping conditions bound every run: total time (including a finalization reserve), steps, commands, retries per operation, global recovery attempts, and repeated no-progress cycles. Any stop still finalizes the trace and keeps `game/`. Before every verification the controller also snapshots the game source into the trace; if a run ends without success, it restores the version that passed the most mandatory criteria and records the decision. The outcome stays FAILED or EXHAUSTED: a better partial version never counts as success. In live runs this machinery handled real provider timeouts, malformed JSON (including empty `{}` replies from GLM-5.3, which the planner now absorbs without spending recovery attempts), runtime errors in generated code, and missing files.

6 Runtime safety and tracing

Runtime (Member 3). Commands run as argument vectors (no shell) inside an OS sandbox (macOS `sandbox-exec`, Linux `bwrap`) that allows writes only to `game/` and the run's private storage, so generated code cannot modify the agent, the specification or the traces. Each command runs in its own process group, which is terminated and confirmed gone after every command, timeout or cancellation. Dependencies install into an isolated environment. Credentials stay in the agent process: commands get a filtered environment and cannot read `agent.env`. A start-up self-check stops the run before `game/` exists if Chromium, the sandbox, a free port or Python is missing; `python -m agent.preflight --probe-model` additionally makes one model call that must return valid JSON, so a wrong key, endpoint or reasoning setting is caught before the run.

Tracing (Member 6). Every controller event is appended to `events.jsonl` and is durable before the controller continues. The trace also holds checkpoints, every verification report, screenshots with a SHA-256 index, model usage per call (role, tokens, latency) and `run_report.md`, a readable timeline with each action's rationale, the frozen criteria, the plans and every verification cycle. `finalize` re-reads what it wrote before reporting the delivery as preserved, and `python -m agent.delivery` audits a finished trace.

7 Evaluation

7.1 Automated tests

226 tests (plus 58 subtests) cover the controller (36), configuration (7), planner (6), runtime and sandbox (22), verifier and harness (26, including real Playwright runs against fixture pages), recovery and evaluation (46), tracing (21), and integration (43, including an offline end-to-end rehearsal with a scripted model and the real sandbox, browser, recovery and tracing), and the final hand-off (19: model probe, source snapshots and best-version restore, rehearsal archiving). A guard test ensures that the agent package contains no game code.

7.2 Live runs with real models

We ran the agent autonomously 39 times on seven specifications (collection, dodge, memory cards, maze, terminal hangman, light switches, turn-based keys) with four models, changing the agent between rounds based on what the traces showed.

Model	Runs	Succeeded	Observations
GLM-5.3 (course model)	16	14	Quality close to Claude Sonnet 5.5; more repair cycles on timing-heavy rules; one failure caused by a harness defect, fixed and re-run successfully
Claude Sonnet 5.5	11	11	Valid scenarios and spec-faithful games in every run
GPT-4.1-mini	8	5	Missed rule details; some generated scenarios could not be passed by a correct game
Qwen3-Coder	4	1	Frequent format errors and unpassable tests

Round 1 used generic checks only, so its successes are weaker evidence; later rounds used spec-derived scenarios. Each specification was run once per model and round, and the agent changed between rounds, so these figures indicate

tendencies rather than rates. GLM-5.3: 9 runs during development and 7 in the final rehearsal (Section 7.3). Full data: `docs/testing/LIVE_MODEL_TESTS.md`.

7.3 The course model: GLM-5.3

A probe before any run revealed a critical behaviour: with no reasoning setting, GLM-5.3 spent all 16 000 output tokens on reasoning and returned **no text** after 81 s, which would have stopped every planning call. Reasoning cannot be disabled for this model, but a reasoning cap makes it answer in about 5 s with valid JSON. The agent now caps reasoning at 8 000 tokens for GLM models. With that setting:

Specification	Outcome	Time	Repairs	Notes
Collection	succeeded	39 s	0	All 7 scenarios passed first time; reviewed against the spec
Terminal hangman	succeeded	17 s	0	Python, process mode; played by hand
Light switches	succeeded	115 s / 202 s	1 / 1	Two runs (before and after a harness fix)
Turn-based keys	succeeded	433 s	1	Caught and fixed "walking into the enemy does not lose"
Dodge	succeeded	453 s	5	Restart and timing rules took several cycles
Memory cards	succeeded	1151 s	8	Mouse only; used almost the whole 20-minute budget
Maze	failed, then succeeded	265 s / 235 s	20 / 6	First run hit a harness defect (function field in the state contract) and empty replies; both fixed

The probes and nine runs cost about US$1.7 in total, which also confirmed that the run fits within a presentation slot.

Final rehearsal. After the last changes, all seven specifications were run once more with the course key (OpenRouter) and a 20-minute budget, with no human intervention:

Specification	Outcome	Time	Steps	Recoveries
Collection	succeeded	48 s	20	0
Memory cards	succeeded	45 s	22	0
Terminal hangman	succeeded	23 s	18	0
Light switches	succeeded	61 s	41	1
Turn-based keys	succeeded	106 s	30	0
Maze	succeeded	334 s	128	5
Dodge	exhausted	1180 s	166	8

The dodge run kept failing timing assertions in model-written scenarios (for example a score that had to be exactly 0 at the start of a check while the real-time loop was already adding points). At the deadline the controller restored the best assessed version (11 mandatory criteria passing) and reported the run as exhausted. The batch cost about US$0.5. Full data: `docs/testing/GLM_REPEAT_20261019.json`.

7.4 Live run at the presentation

To be completed after the presentation. Specification received, outcome and reason, elapsed time, number of verification cycles and repairs, the most important failure and how the agent handled it, and where the evidence is in `traces/<run_id>/`.

8 Current limitations

- **Model-written tests can be wrong.** The scenarios come from the same model that writes the game. Filtering removes known bad patterns, but a scenario that a correct game cannot pass still leads to repeated repairs until a budget runs out. Stronger models write far fewer such tests.
- **Real-time games are the weakest case.** Scenarios assert on wall-clock behaviour, so timers and continuous scoring needed up to eight repair cycles, and the dodge game exhausted its budget in the final rehearsal.
- **Non-browser games are verified less deeply.** Terminal games are played through standard input, but spec-derived scenarios apply to browser games only, and pygame input is checked statically.
- **Demo machine not yet rehearsed.** The course key (OpenRouter) has been tested, and `--probe-model` catches configuration errors, but the final presentation machine still needs a full run.
- **Small evaluation.** Seven specifications and at most two GLM-5.3 runs per specification; results are indicative.
- **Platform.** The command sandbox requires macOS or Linux with bubblewrap; there is no resume after a hard interruption.

9 Next planned improvement

Deterministic time for real-time games. The two improvements planned earlier, keeping the best verified version and a model connectivity check, are now implemented (Sections 5 and 6). The remaining weak point is real-time games, whose scenarios currently race the game loop. The state contract will require a `pause` hook and a `step(ms)` hook that advances the simulation by a fixed amount, and the harness will run timing assertions on a paused game advanced step by step. Assertions such as "the score is still 0" or "an obstacle moves down" then become exact and repeatable instead of depending on frame timing, which targets the failure seen in the dodge rehearsal.

A smaller follow-up: **standard setup hooks** per game type (place an item next to the player, open a grid cell, set a score) to reduce unpassable model-written scenarios.

10 Use of AI assistance

The team used Codex (OpenAI) and Claude Code (Anthropic) as coding assistants during development. They were used to integrate the members' branches, diagnose and fix cross-module defects found in testing (for example the verifier/runtime data mismatch, the interaction check that missed moving players, and macOS process-group cleanup), write additional tests. Members designed and implemented their own modules (Section 11) and reviewed the integration changes to their modules before the merge into

the main branch; later changes were made by Member 1 with the same assistance and are recorded in the Git history.

At run time the agent itself calls only the course-designated model through LiteLLM. It does not invoke any existing coding agent, and it contains no pre-written game code: every game is generated during the run, and a test enforces that the agent package ships no game files.

11 Member contributions

Member	Contribution
1 · ZHOU YU	Team interface contract; finite-state controller, budgets and success gate; configuration and CLI. **Integration:** merged the six member branches into `main`, built the production factory, fixed cross-module defects, and reviewed and merged the final hand-off (PR #7). **Hardening:** start-up self-check, `agent.env` with credential isolation, retry backoff and configurable run limits. **Verification and model support** (with Members 4 and 2): spec-derived acceptance scenarios and the state contract, process-mode checks for terminal and pygame games, the GLM-5.3 reasoning cap and empty-reply handling. **Evaluation:** 32 live runs with four models over five rounds, and their analysis. **Documentation:** README and artwork, testing guides, this report and the presentation slides. Work after the member merge used the AI assistance described in Section 10.
2 · JIN ZIYI	Planning module: LiteLLM model client, structured output schemas, plan and decision parsing and validation, planning and decision prompts, launch-command and plan-coverage validation, planner tests.
3 · DAI MINFEI	Tool runtime: workspace files and unified-diff patches, command execution, OS sandbox, process-group cleanup, isolated dependency installation, workspace revisions, runtime tests and test reports.
4 · ZHOU SIHAN	Verification module: `GameVerifier`, acceptance criteria, evidence ledger and criterion evaluation, the Playwright game harness, verifier tests and design notes.
5 · LIN SIYUAN	Recovery module: failure classification and recovery policy; offline evaluation specifications and run-record report; recovery and recovery-verifier integration tests; model probe in the self-check, source snapshots and best-version restore; final GLM-5.3 rehearsal and audit of the historical run archive.
6 · LI JIAKUN	Tracing and delivery: durable trace recorder, checkpoints, verification screenshot archiving with integrity checks, delivery audit including missing-evidence detection, tracing tests.

A Appendix: running the agent

Step	Command
Install (once)	`pip install -e ".[test]"` then `python -m playwright install chromium-headless-shell`
Configure	`agent.env` next to the specification: `AGENT_MODEL=openrouter/z-ai/glm-5.3`, `OPENROUTER_API_KEY=…`, optional `AGENT_MAX_RUNTIME`
Check the machine	`python -m agent.preflight`, then `python -m agent.preflight --probe-model` (one billed call)
Run	`python -m agent game_design_prompt.md`
Start the game	the command in `game/README.md`, normally `cd game && python3 -m http.server 8000`
Inspect the run	`traces/<run_id>/run_report.md`; audit with `python -m agent.delivery traces/<run_id>`